

MindSpore Transformers 大模型系列课程

MindSpore Transformers & Megatron-LM训练精度对比

主讲人：Ben

昇思MindSpore研发工程师

MindSpore Transformers SIG Maintainer

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《MindSpore Transformers LLM训练迁移与实践》

《MindSpore Transformers LLM推理迁移与实践》

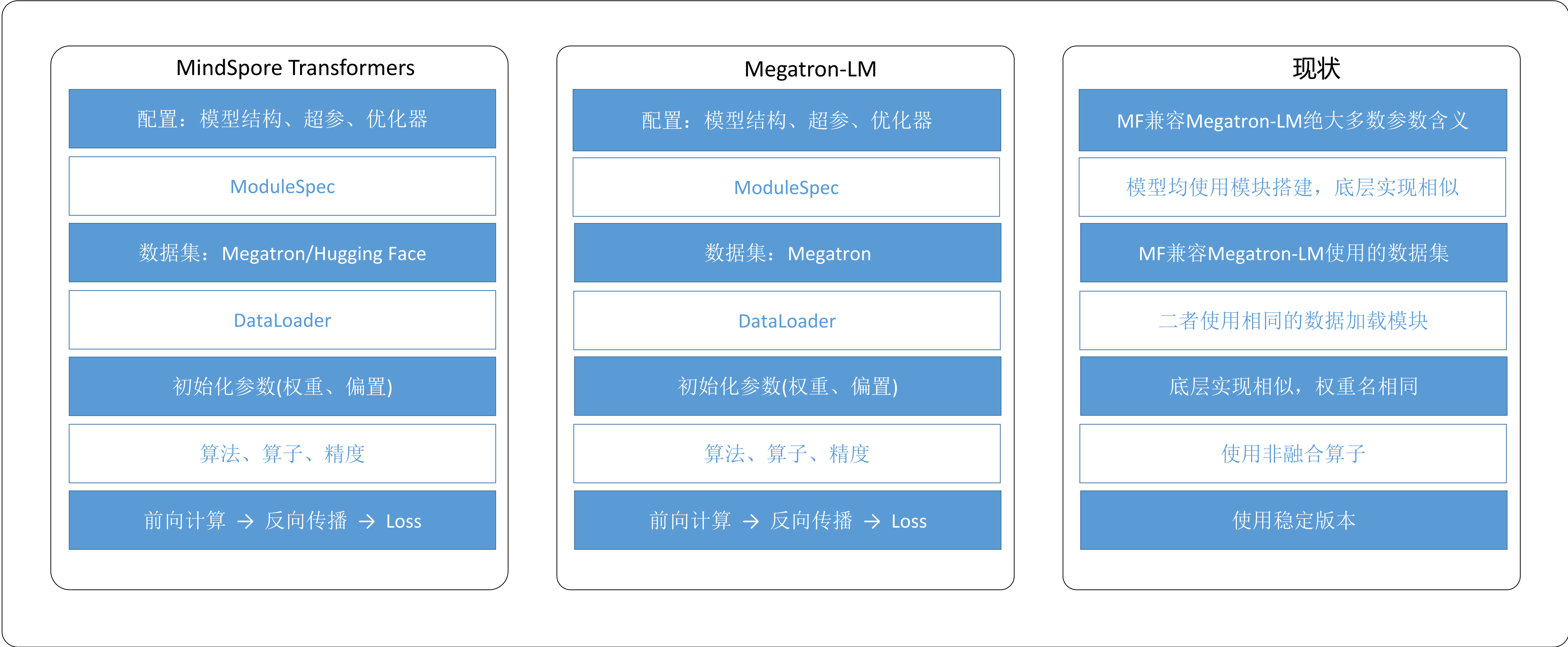
《MindSpore Transformers训练性能调优》

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- 2. 训练精度比对流程详述**
- 3. 训练精度比对问题定位样例**

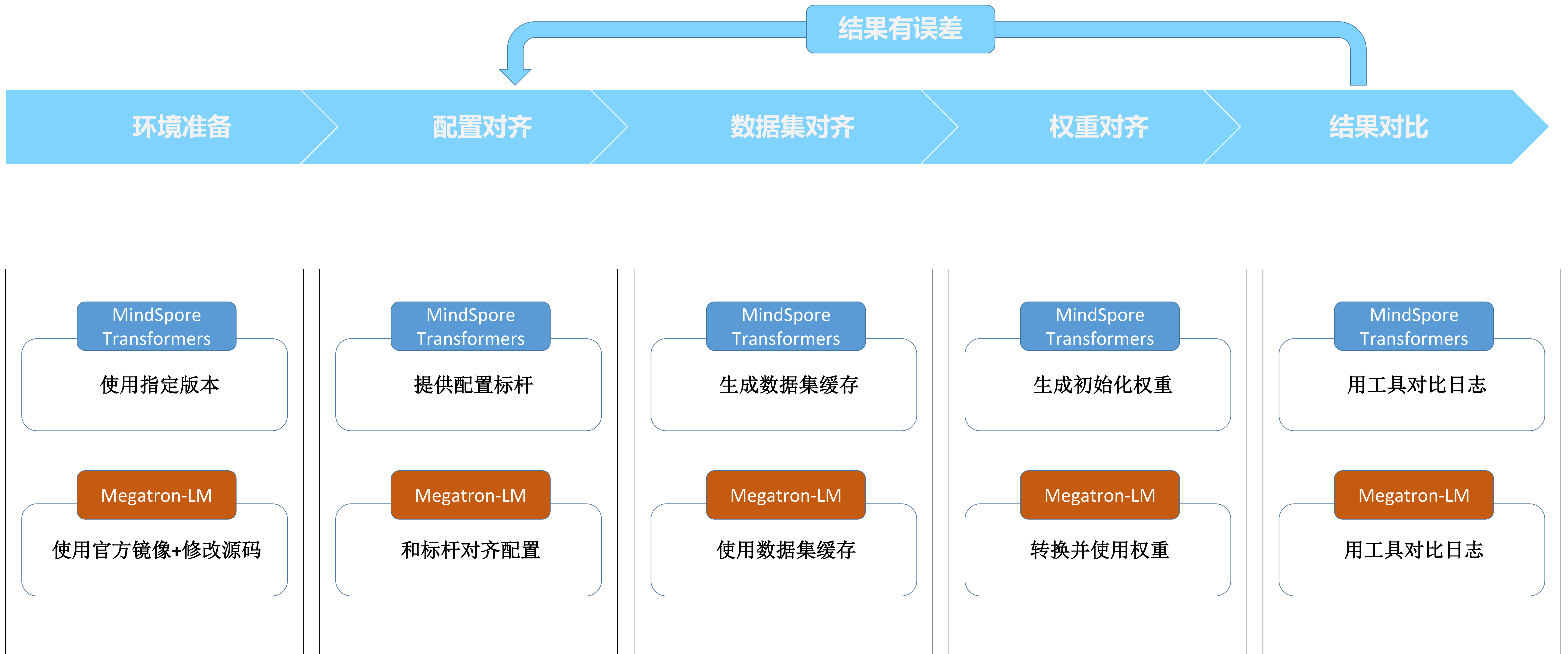
1. 训练精度对比概念介绍

训练流程



2.训练精度对比流程详述

训练精度对比流程图



环境准备阶段



NPU	版本	GPU	版本
MindSpore Transformers	>=1.6.0	Megatron-LM	core_r0.12.0 commit_id: d60d187a8138e745f5580b3fb20158479e9abadf
Python	>=3.10		
MindSpore	>=2.7.0		
NumPy	1.26.4		
CANN	>= 8.2.RC1		

	NPU	GPU
获取途径	参照 官方文档 获取搭建方法	官方文档 获取镜像路径

环境准备阶段

环境准备

配置对齐

数据集对齐

权重对齐

结果对比

进入Megatron-LM目录



应用补丁文件或者手动修改

修改点

修改点一

文件: megatron/core/models/gpt/gpt_layer_specs.py

修改内容: 在 65 行增加以下代码

```
1 from megatron.core.transformer.torch_norm import WrappedTorchNorm
2 LNImpl = WrappedTorchNorm
```

原因: 将 TENorm 替换成 torch Norm, 避免融合算子的干扰

修改点二

文件: megatron/core/transformer/multi_latent_attention.py

修改内容: 在 114-116 行将以下代码注释掉

```
1 #k_channels=self.q_head_dim,
2 #v_channels=self.config.v_head_dim,
3 #cp_comm_type=cp_comm_type,
```

原因: Torch 的 flash_attention 不接受这三个入参, 需要注释掉

修改点三

文件: megatron/core/transformer/multi_token_prediction.py

修改内容: 在 60 行加入以下代码

```
1 +from megatron.core.transformer.torch_norm import WrappedTorchNorm
2 +LNImpl = WrappedTorchNorm
```

原因: 将 TENorm 替换成 torch Norm, 避免融合算子的干扰

修改点四

文件: megatron/training/checkpointing.py

修改内容: 在 1373 行加入以下代码

```
1 strict = False
```

原因: strict=False 可以不严格加载权重, 可以缺失一些 key 值, 因为我们不提供优化器的初始化权重, 需要不严格加载

配置对齐阶段

环境准备	配置对齐	数据集对齐	权重对齐	结果对比
MindSpore Transformers	含义	Megatron-LM	含义	备注
use_legacy	是否使用 mcore 模型	use-legacy-model 和 use-mcore-model组合	是否使用 mcore 模型	use_legacy:True等价于--use-legacy-model, use_legacy:False等价于--use-mcore-model
vocab_size	词表大小	vocab-size	词表大小	
hidden_size	隐藏层大小，隐藏状态的维度	hidden-size	隐藏层大小，隐藏状态的维度	
intermediate_size或ffn_hidden_size	前馈网络隐藏层大小	ffn-hidden-size	前馈网络隐藏层大小	MindSpore Transformers支持多种输入且会相互覆盖

环境准备

配置对齐

数据集对齐

权重对齐

结果对比

1. 数据加载模块会根据随机生成的哈希值和所需数据量去读取数据集；
2. 数据量的配置在MF通过sizes配置，在Megatron通过global-batch-size和train-iters的乘积控制，需要控制这两者相等；
3. 哈希值可以通过先进行一次训练生成，得到的cashe文件会包含生成的哈希值，然后复制到另一个框架使用

```
train_dataset: &train_dataset
data_loader:
  type: BlendedMegatronDatasetDataLoader
  datasets_type: "GPTDataset"
  sizes:
    - 4000 # Number of samples in the training set
    - 0    # Number of samples in the test set (currently unsupported)
    - 0    # Number of samples in the evaluation set (currently unsupported)
  config:
    seed: 1234 # Random seed for data sampling
    split: "1, 0, 0" # Proportions for training, test, and evaluation sets (test/eval currently unsupported)
```

```
DATA_ARGS=(
  --tokenizer-type HuggingFaceTokenizer
  --tokenizer-model ${TOKENIZER_MODEL}
  --data-path $DATA_PATH
  --split 1,0,0
  --seed 1234
)
```

```
TRAINING_ARGS=(
  --micro-batch-size 1
  --global-batch-size 4
  --train-iters 1000
```


权重对齐阶段

环境准备

配置对齐

数据集对齐

权重对齐

结果对比

- 1.在NPU生成初始化的权重0(数据类型是FP32，因为优化器的权重是FP32)
- 2.将权重0转换成数据类型BF16的权重1(Megatron-LM加载权重使用的精度是BF16)
- 3.将权重1合并成权重2(用于后续转换成Megatron-LM的分布式权重)
- 4.由于MF对性能做了优化，所以权重2的部分权重排布需要进行额外处理才可以转换
- 5.将权重2转换成排布适合Megatron的权重3
- 6.将权重3转换成Megatron权重4
- 7.MF使用合并权重后的权重2进行训练，Megatron使用最终转换的权重4进行训练

权重对齐场景演练

生成权重

```
output_dir: './output'
# checkpoints setting
load_checkpoint: '' # 'path/to/unified_safe'
load_ckpt_format: 'safetensors'
auto_trans_ckpt: False # If true, auto transform load_checkpoint to load in distributed model
```

```
# callbacks
callbacks:
- type: MFLossMonitor
- type: CheckpointMonitor
  prefix: "qwen3_moe"
  save_checkpoint_steps: 100
  keep_checkpoint_max: 2
  integrated_save: False
  async_save: False
  checkpoint_format: "safetensors"
```

```
2025-11-06 11:05:42,524 - /output/log[mindformers/core/callback/callback.py:1435] -
INFO - .....Saving ckpt.....
2025-11-06 11:05:42,524 - /output/log[mindformers/core/callback/callback.py:1494] -
INFO - global_batch_size: 16
2025-11-06 11:05:42,524 - /output/log[mindformers/core/callback/callback.py:1495] -
INFO - epoch_num: 100
2025-11-06 11:05:42,525 - /output/log[mindformers/core/callback/callback.py:1496] -
INFO - step_num: 100
2025-11-06 11:05:42,525 - /output/log[mindformers/core/callback/callback.py:1497] -
INFO - global_step: 100
2025-11-06 11:05:49,051 - /output/log[mindformers/core/callback/callback.py:1350] -
INFO - Finish saving ckpt of epoch 100 step 1 using 6.501 seconds
2025-11-06 11:05:50,832 - /output/log[mindformers/core/callback/callback.py:374] -
WARNING - pipeline stages: 8 > 1, the loss on the last card is valid.
2025-11-06 11:05:50,835 - /output/log[mindformers/core/callback/callback.py:544] -
INFO - { Epoch:[ 1/ 1], step:[ 101/ 2000], loss: 0.000000, lm_loss: 0.000000, per_step_time: 1778ms, lr: 1e-05, overflow co
nd: False, loss scale: 1.0, global_norm: [1.7036879]}
2025-11-06 11:05:50,836 - /output/log[mindformers/core/callback/callback.py:560] -
INFO - 5.1% |■■ | 1.12455 samples/s/p 0:56:17 }
```

```
total 40
drwxr-x---. 10 root root 4096 Nov  6 11:01 ./
drwx-----. 6 root root 4096 Nov  6 10:53 ../
drwx-----. 2 root root 4096 Nov  6 11:05 rank_0/
drwx-----. 2 root root 4096 Nov  6 11:05 rank_1/
drwx-----. 2 root root 4096 Nov  6 11:05 rank_2/
drwx-----. 2 root root 4096 Nov  6 11:05 rank_3/
drwx-----. 2 root root 4096 Nov  6 11:05 rank_4/
drwx-----. 2 root root 4096 Nov  6 11:05 rank_5/
drwx-----. 2 root root 4096 Nov  6 11:05 rank_6/
drwx-----. 2 root root 4096 Nov  6 11:05 rank_7/
```

- 到达指定步数后，结束训练
- 在指定目录生成分布式权重

转换数据类型

```
Processing output/checkpoint/rank_7/qwen3_moe_rank_7-100_1.safetensors: key='aux_loss' Float32 => BFloat16
Processing output/checkpoint/rank_7/qwen3_moe_rank_7-100_1.safetensors: key='decoder.final_layernorm.weight' Float32 => BFloat16
Processing output/checkpoint/rank_7/qwen3_moe_rank_7-100_1.safetensors: key='decoder.layers.7.input_layernorm.weight' Float32 => BFloat16
Processing output/checkpoint/rank_7/qwen3_moe_rank_7-100_1.safetensors: key='decoder.layers.7.mlp.experts.weight1' Float32 => BFloat16
Processing output/checkpoint/rank_7/qwen3_moe_rank_7-100_1.safetensors: key='decoder.layers.7.mlp.experts.weight2' Float32 => BFloat16
Processing output/checkpoint/rank_7/qwen3_moe_rank_7-100_1.safetensors: key='decoder.layers.7.mlp.router.weight' Float32 => BFloat16
Processing output/checkpoint/rank_7/qwen3_moe_rank_7-100_1.safetensors: key='decoder.layers.7.pre_mlp_layernorm.weight' Float32 => BFloat16
Processing output/checkpoint/rank_7/qwen3_moe_rank_7-100_1.safetensors: key='decoder.layers.7.self_attention.k_layernorm.weight' Float32 => BFloat16
Processing output/checkpoint/rank_7/qwen3_moe_rank_7-100_1.safetensors: key='decoder.layers.7.self_attention.linear_proj.weight' Float32 => BFloat16
Processing output/checkpoint/rank_7/qwen3_moe_rank_7-100_1.safetensors: key='decoder.layers.7.self_attention.linear_qkv.weight' Float32 => BFloat16
Processing output/checkpoint/rank_7/qwen3_moe_rank_7-100_1.safetensors: key='decoder.layers.7.self_attention.q_layernorm.weight' Float32 => BFloat16
Processing output/checkpoint/rank_7/qwen3_moe_rank_7-100_1.safetensors: key='lm_loss' Float32 => BFloat16
Processing output/checkpoint/rank_7/qwen3_moe_rank_7-100_1.safetensors: key='mtp_loss' Float32 => BFloat16
Processing output/checkpoint/rank_7/qwen3_moe_rank_7-100_1.safetensors: key='output_layer.weight' Float32 => BFloat16
Saved converted checkpoint to output/checkpoint/rank_7/qwen3_moe_rank_7-100_1.safetensors
Completed processing output/checkpoint/rank_7/qwen3_moe_rank_7-100_1.safetensors and cleared memory
```

- convert2bf16.py
- 将分布式权重的数据类型修改为BFloat16
- <10G, 半分钟

权重对齐场景演练

合并权重

```
python unified_safetensors.py \  
  --mindspore_ckpt_dir /path/checkpoint \  
  --src_strategy_dirs /path/src_strategy_dirs \  
  --output_dir /path/src_strategy_dirs \  
  --file_suffix "1_1" \  
  --format "safetensors" \  
  --has_redundancy False \  

```

```
2025-11-06 11:32:19,735 - mindformers/data/test/mindformers/output/log[mindformers/toolkit/safetensors/unified_safetensors.py:  
76] - INFO - [2025-11-06 11:32:19 args config:]  
2025-11-06 11:32:19,735 - mindformers/data/test/mindformers/output/log[mindformers/toolkit/safetensors/unified_safetensors.py:  
99] - INFO - file suffix=100_1  
2025-11-06 11:32:19,735 - mindformers/data/test/mindformers/output/log[mindformers/toolkit/safetensors/unified_safetensors.py:  
99] - INFO - filter_out_param_prefix=None  
2025-11-06 11:32:19,736 - mindformers/data/test/mindformers/output/log[mindformers/toolkit/safetensors/unified_safetensors.py:  
99] - INFO - format=safetensors  
2025-11-06 11:32:19,736 - mindformers/data/test/mindformers/output/log[mindformers/toolkit/safetensors/unified_safetensors.py:  
99] - INFO - has_redundancy=False  
2025-11-06 11:32:19,736 - mindformers/data/test/mindformers/output/log[mindformers/toolkit/safetensors/unified_safetensors.py:  
99] - INFO - max_process_num=64  
2025-11-06 11:32:19,736 - mindformers/data/test/mindformers/output/log[mindformers/toolkit/safetensors/unified_safetensors.py:  
99] - INFO - mindspore_ckpt_dir=output/checkpoint/  
2025-11-06 11:32:19,736 - mindformers/data/test/mindformers/output/log[mindformers/toolkit/safetensors/unified_safetensors.py:  
99] - INFO - output_dir=output/100_1 ckpt_convert  
2025-11-06 11:32:19,736 - mindformers/data/test/mindformers/output/log[mindformers/toolkit/safetensors/unified_safetensors.py:  
99] - INFO - src_strategy_dirs=output/strategy/  
2025-11-06 11:32:19,736 - mindformers/data/test/mindformers/output/log[mindformers/toolkit/safetensors/unified_safetensors.py:  
76] - INFO - [2025-11-06 11:32:19 Task start...]  
2025-11-06 11:32:19,737 - mindformers/data/test/mindformers/output/log[mindformers/toolkit/safetensors/unified_safetensors.py:  
76] - INFO - [2025-11-06 11:32:19 Start merge strategy]  
2025-11-06 11:32:19,799 - mindformers/data/test/mindformers/output/log[mindformers/toolkit/safetensors/unified_safetensors.py:  
76] - INFO - [2025-11-06 11:32:19 Merge strategy completed]  
2025-11-06 11:32:19,800 - mindformers/data/test/mindformers/output/log[mindformers/toolkit/safetensors/unified_safetensors.py:  
76] - INFO - [2025-11-06 11:32:19 Start merge safetensor]  
[WARNING] ME(1033688:281473178895968,MainProcess):2025-11-06-11:32:19.841.000 [mindspore/parallel/transform_safetensors.py:116  
8] For 'unified_safetensors', the safetensors file is non-duplicated, but merge_with_redundancy is set to False.  
2025-11-06 11:32:26,072 - mindformers/data/test/mindformers/output/log[mindformers/toolkit/safetensors/unified_safetensors.py:  
76] - INFO - [2025-11-06 11:32:26 Merge safetensor completed]  
2025-11-06 11:32:26,073 - mindformers/data/test/mindformers/output/log[mindformers/toolkit/safetensors/unified_safetensors.py:  
100] - INFO - merged safetensor path: output/100_1 ckpt_convert/unified safe  
2025-11-06 11:32:26,073 - mindformers/data/test/mindformers/output/log[mindformers/toolkit/safetensors/unified_safetensors.py:  
76] - INFO - [2025-11-06 11:32:26 Task completed time:]  
2025-11-06 11:32:26,073 - mindformers/data/test/mindformers/output/log[mindformers/toolkit/safetensors/unified_safetensors.py:  
76] - INFO - [2025-11-06 11:32:26 Task total cost time: 6.336233615875244s]
```

- toolkit/safetensors
- 在指定目录生成合并权重
- <10G, 数秒

转换权重排布

```
moe_ffn_hidden_size: 768  
num_local_experts: 64  
hidden_size: 1024  
decoder.layers.1.mlp.experts.weight1  
(65536, 1536)  
(1024, 98304)
```

- reorganize_ckpt.py
- 在指定目录生成转换排布后的权重
- <10G，几分钟

权重对齐场景演练

转换权重

- convert2megatron.py
- <10G, 几分钟

```
decoder.final_layernorm.weight
torch.Size([1024])
decoder.layers.0.input_layernorm.weight
torch.Size([1024])
decoder.layers.0.mlp.experts.weight1
torch.Size([1024, 98304])
decoder.layers.0.mlp.experts.weight2
torch.Size([49152, 1024])
decoder.layers.0.mlp.router.weight
torch.Size([64, 1024])
decoder.layers.0.pre_mlp_layernorm.weight
torch.Size([1024])
decoder.layers.0.self_attention.k_layernorm.weight
torch.Size([128])
decoder.layers.0.self_attention.linear_proj.weight
torch.Size([1024, 4096])
decoder.layers.0.self_attention.linear_qkv.weight
torch.Size([5120, 1024])
decoder.layers.0.self_attention.q_layernorm.weight
torch.Size([128])
output_layer.weight
torch.Size([129280, 1024])
-----
pp id: 0, tp id: 0
qwen3_moe/iter_0000001/mp_rank_00_000/model_optim_rng.pt
pp id: 1, tp id: 0
qwen3_moe/iter_0000001/mp_rank_00_001/model_optim_rng.pt
pp id: 2, tp id: 0
qwen3_moe/iter_0000001/mp_rank_00_002/model_optim_rng.pt
pp id: 3, tp id: 0
qwen3_moe/iter_0000001/mp_rank_00_003/model_optim_rng.pt
pp id: 4, tp id: 0
qwen3_moe/iter_0000001/mp_rank_00_004/model_optim_rng.pt
pp id: 5, tp id: 0
qwen3_moe/iter_0000001/mp_rank_00_005/model_optim_rng.pt
pp id: 6, tp id: 0
qwen3_moe/iter_0000001/mp_rank_00_006/model_optim_rng.pt
pp id: 7, tp id: 0
qwen3_moe/iter_0000001/mp_rank_00_007/model_optim_rng.pt
```

加载权重

记得修改yaml文件

```
load_checkpoint: 'path/to/unified_safe'
load_ckpt_format: 'safetensors'
auto_trans_ckpt: True # If true, auto transform load_checkpoint to load in distributed model
```

```
output_dir: './output'
# checkpoints setting
load_checkpoint: '' # 'path/to/unified_safe'
load_ckpt_format: 'safetensors'
auto_trans_ckpt: False # If true, auto transform load_checkpoint to load in distributed model
```

```
# callbacks
callbacks:
- type: MFLossMonitor
- type: CheckpointMonitor
  prefix: "qwen3_moe"
  save_checkpoint_steps: 100
  keep_checkpoint_max: 2
  integrated_save: False
  async_save: False
  checkpoint_format: "safetensors"
```

```
# callbacks
callbacks:
- type: MFLossMonitor
  #- type: CheckpointMonitor
  #prefix: "qwen3_moe"
  #save_checkpoint_steps: 100
  #keep_checkpoint_max: 2
  #integrated_save: False
  #async_save: False
  #checkpoint_format: "safetensors"
```




MindSpore Transformers	Megatron-LM	含义
epoch 与 step 的组合	iteration	表示训练过程中的全局迭代次数。MindSpore Transformers 通常以 (epoch, step) 表示当前训练位置，而 Megatron-LM 使用单一的 iteration 表示。两者关系为：iteration = (epoch - 1) * steps_per_epoch + step
lm_loss	lm loss	语言模型主要训练损失，精度对比核心指标
load_balancing_loss	load_balancing_loss	负载均衡损失，MoE模型精度对比重要指标
lr	learning rate	学习率，精度对比参考指标
global norm	grad norm	全局梯度范数，精度对比参考指标

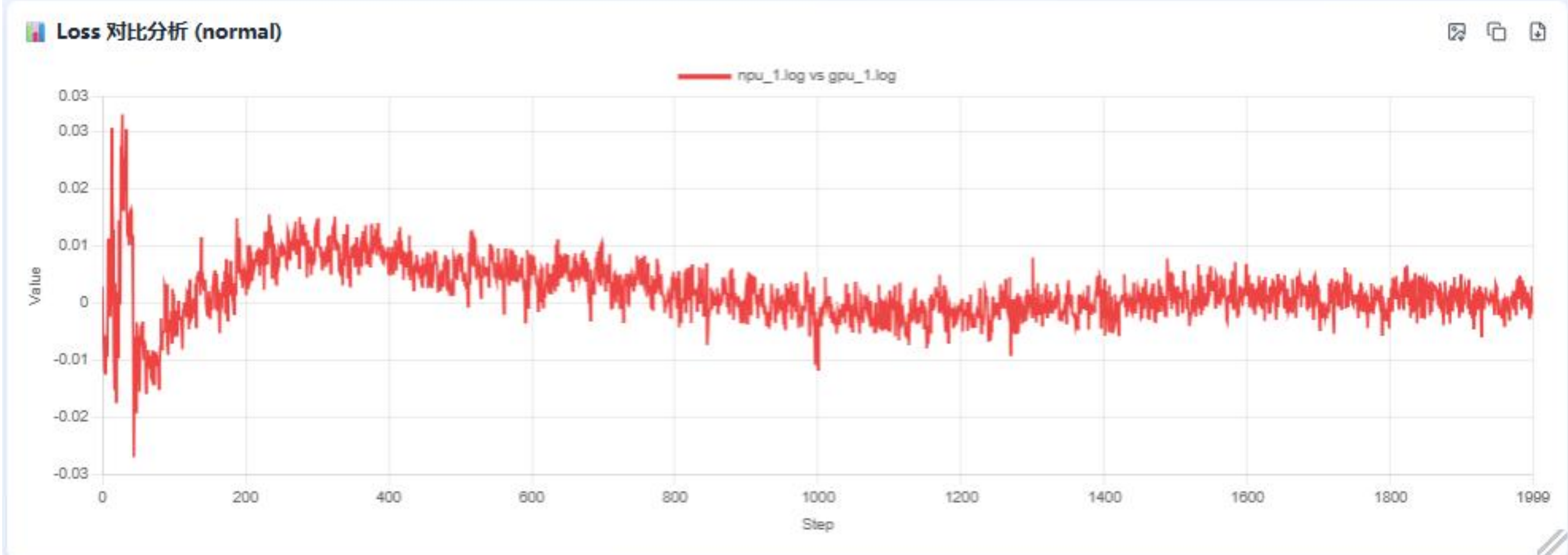
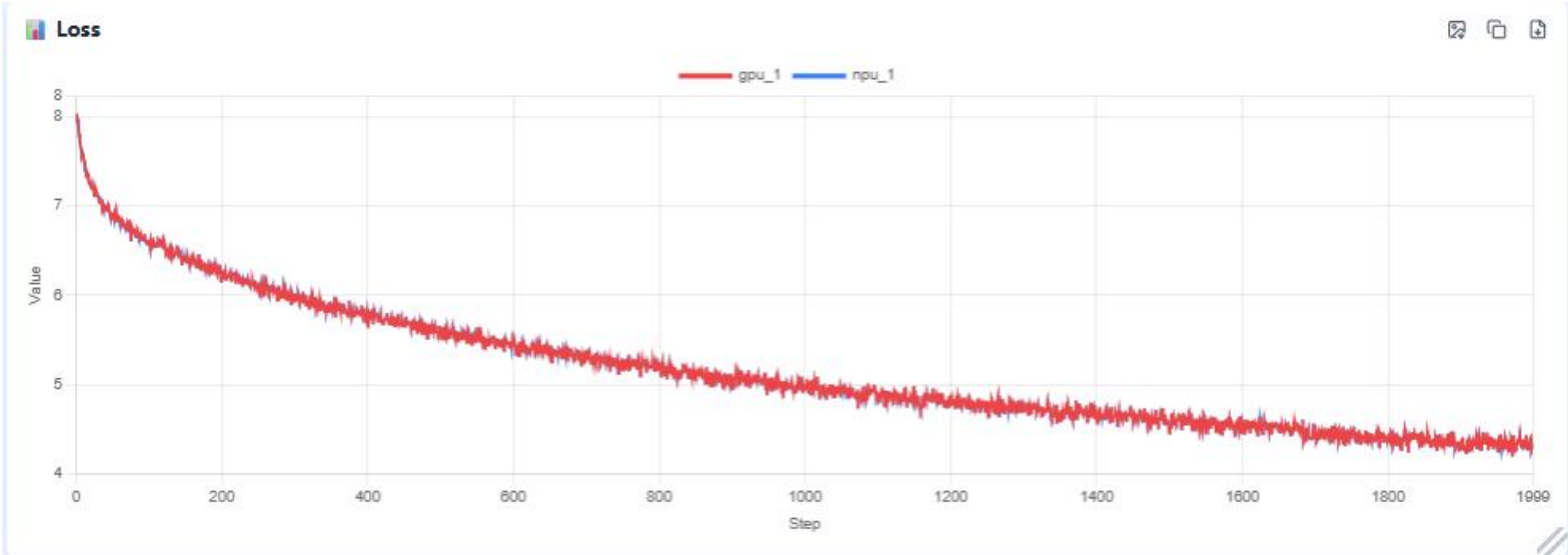
- Loss误差标准
- 首loss绝对误差小于0.005

•平均相对误差小于1%，或平均绝对误差小于0.01

2.训练精度比对问题定位样例

结果比对场景演练

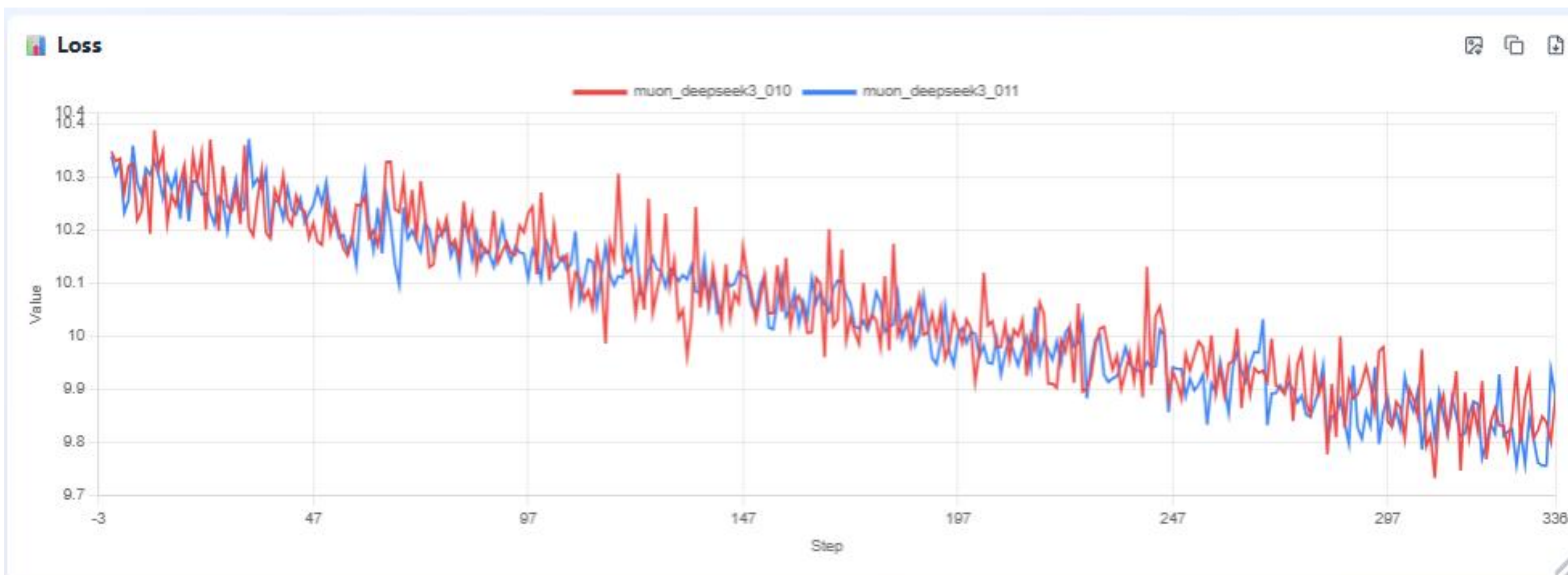
log.javazero.top



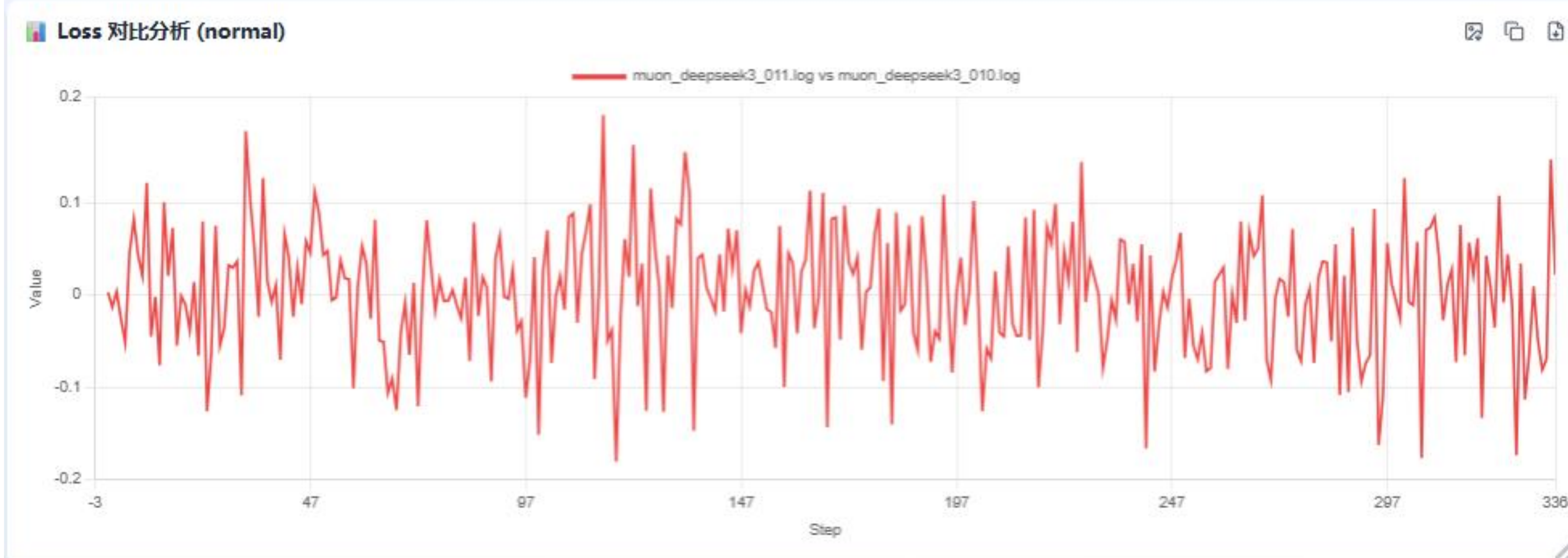
Loss 差值统计					
文件对	平均误差 (normal)	平均误差 (absolute)	最大绝对误差	平均相对误差 (absolute)	最大相对误差
npu_1.log vs gpu_1.log	-0.001040	0.003884	0.030092	0.000701	0.004148

整体误差小于1%

结果比对场景演练



- 拟合程度不高
- 数值范围接近



Loss 差值统计

文件对

muon_deepseek3_011.log vs muon_deepseek3_010.log

平均误差 (normal)

-0.011909

平均误差 (absolute)

0.052942

最大绝对误差

0.192904

平均相对误差 (absolute)

0.005299

最大相对误差

0.018971

拟合程度不高

查看日志

```
----- arguments -----
account_for_embedding_in_pipeline_split ..... False
account_for_loss_in_pipeline_split ..... False
accumulate_allreduce_grads_in_fp32 ..... True
adam_beta1 ..... 0.9
adam_beta2 ..... 0.95
adam_eps ..... 1e-08
```

```
padded_vocab_size ..... 151936
```

```
^M100.0%|#####| [00:00:00<00:00:00]
2025-10-28 09:22:29.420 - mindformers/data/z30061521/mindformers/output/log[mindformers/utils/load_checkpoint_utils.py:437] -
INFO - load checkpoint unified: output/10_1_ckpt_convert/unified_safe
{'adapter_id': None,
 'auto_trans_ckpt': True,
 'balanced_load': False,
 'callbacks': [{'type': 'MFLossMonitor'}],
```

```
'vocab_size': 129280
```

pretrain_gpt.py

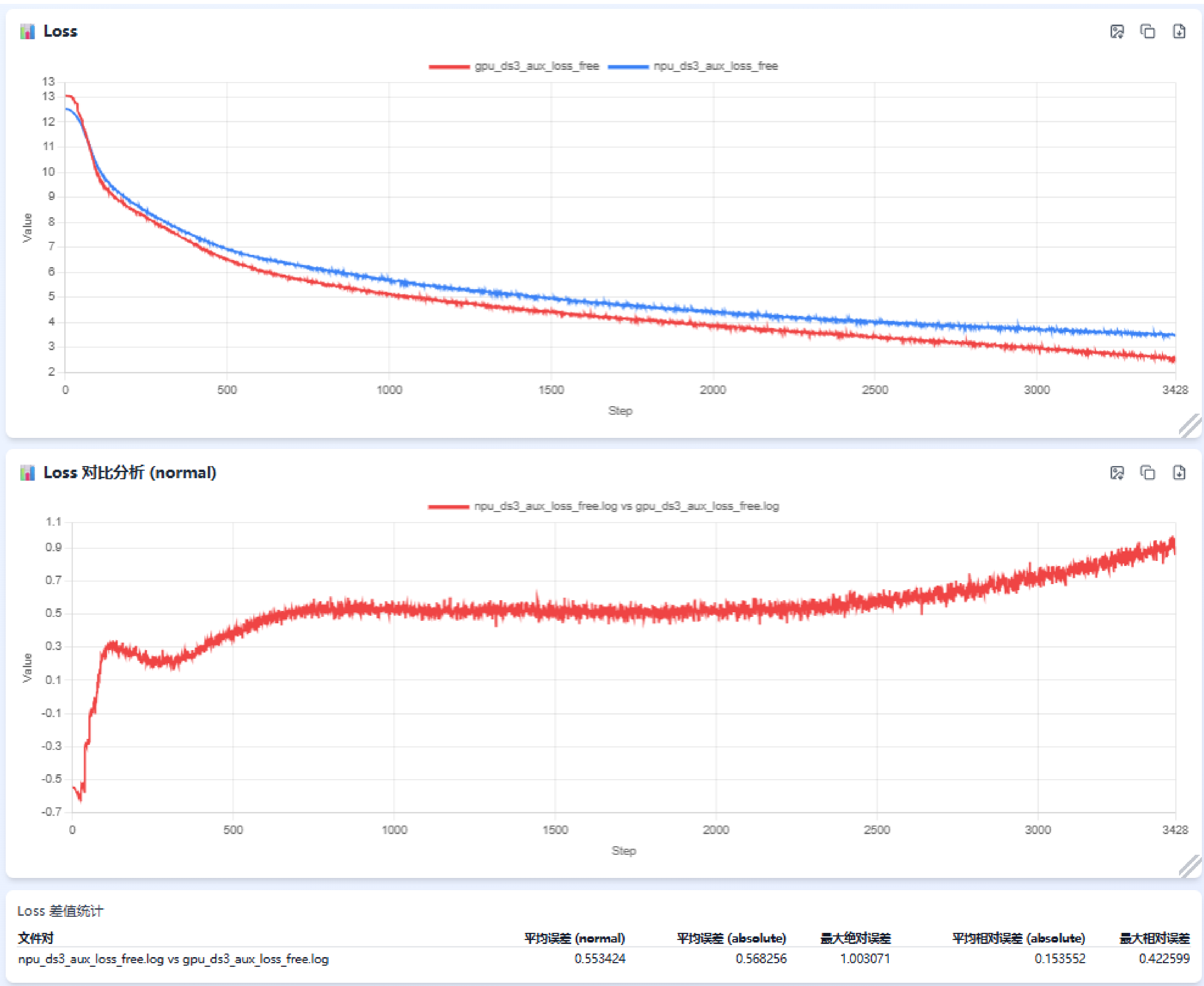
修改文件

```
def model_provider(pre_process=True, post_process=True) -> Union[GPTModel, megatron.legacy.model.GPTModel]:
```

```
    args.padded_vocab_size = 151936
    model = GPTModel(
        config=config,
        transformer_layer_spec=transformer_layer_spec,
        vocab_size=args.padded_vocab_size,
        max_sequence_length=args.max_position_embeddings,
        pre_process=pre_process,
        post_process=post_process,
        fp16_lm_cross_entropy=args.fp16_lm_cross_entropy,
        parallel_output=True,
        share_embeddings_and_output_weights=not args.untie_embeddings_and_output_weights,
        position_embedding_type=args.position_embedding_type,
        rotary_percent=args.rotary_percent,
        rotary_base=args.rotary_base
    )

    return model
```


结果比对场景演练



- 首loss不一致
- 趋势接近

首Loss不一致

MindSpore Transformers:
mindformers/dataset/blended_datasets/gpt_dataset.py

Megatron-LM:
megatron/core/datasets/gpt_dataset.py

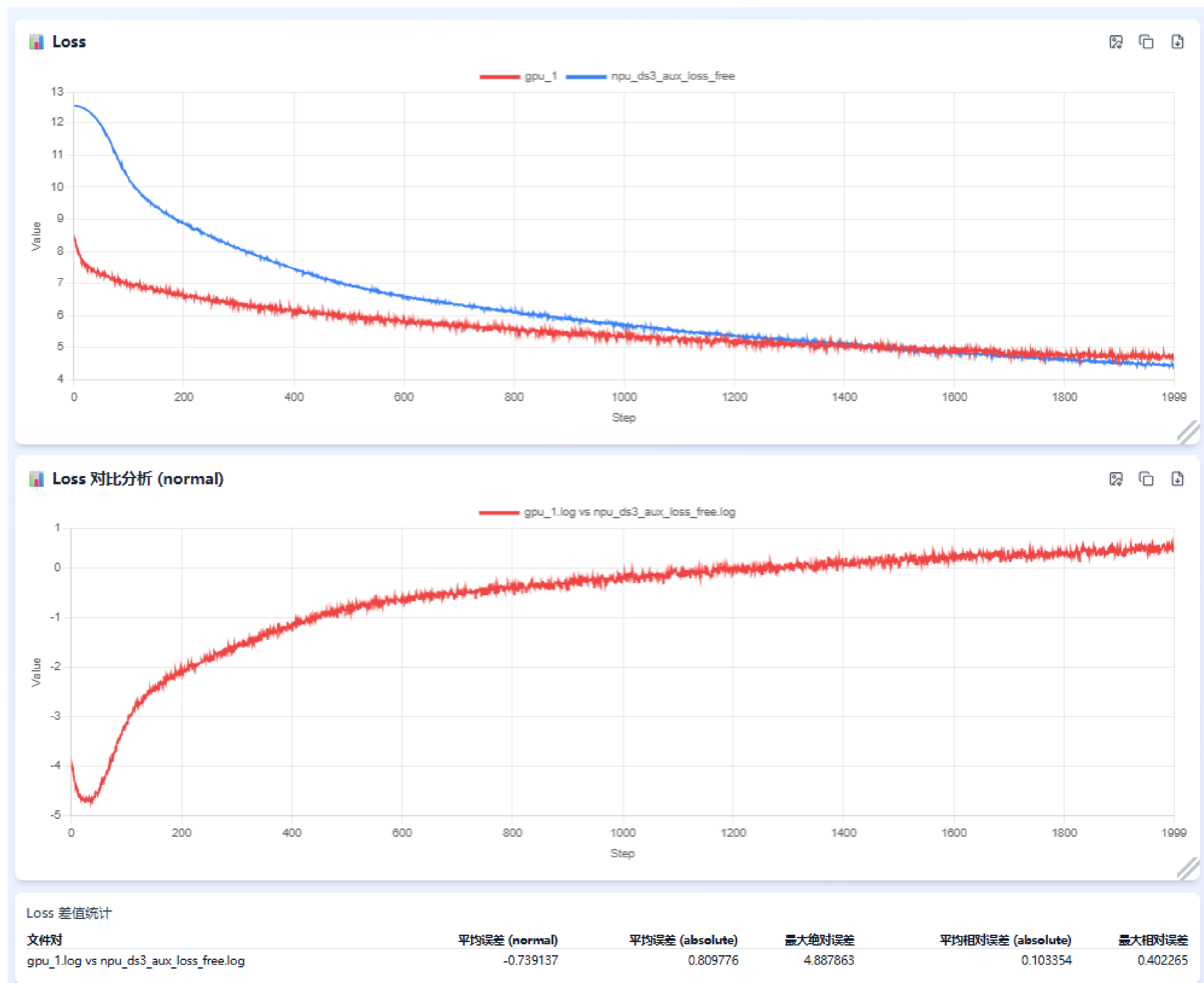
```
class GPTDataset(MegatronDataset):
```

```
def __getitem__(self, idx: Optional[int]) -> Dict[str, numpy.ndarray]:
    """Abstract method implementation

    Args:
        idx (Optional[int]): The index into the dataset

    Returns:
        Dict[str, numpy.ndarray]: The sample information wrapped in a dictionary
    """
    if idx is None:
        # Batch padding sequence so the index does not matter
        text, _ = self._query_document_sample_shuffle_indices(0)
    else:
        text, _ = self._query_document_sample_shuffle_indices(idx)
    print(idx, text)
```


结果比对场景演练



- 首loss不一致
- 趋势不一致

首Loss不一致、趋势不一致

- 权重加载

INFO -Start load checkpoint from safetensors.....

INFO - Load checkpoint from 100_1_ckpt_convert/unified_safe.

successfully loaded checkpoint from ../qwen3_moe/ [t 1/1, p 1/8] at iteration 0

- 权重查看

MindSpore Transformers:
import mindspore as ms
ms.load_checkpoint

Megatron-LM:
import torch
torch.load

谢谢观看！

MindSpore Transformers & Megatron-LM训练精度对比

MindSpore Transformers 大模型系列课程